Answer Key & Teacher Notes

Session 1 — Absolute Value as Distance · Session 2 — Absolute Value as Total Distance

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Absolute Value as Distance

TALK PROMPTS

- Why can two different temperatures have exactly the same absolute value?
- Why would a bank care about the absolute value of a negative balance?

PARTNER TASK

Two Accounts. One partner plots −45 and 12 on a number line; the other writes both absolute values and explains which balance is closer to zero. Switch roles with −30 and 25: which number is greater, and which is farther from 0?

WATCH FOR

- Writing a negative absolute value ($|-5.2| = -5.2$).
- Thinking a negative number is closer to 0 because it is "smaller."
- Mixing up "greater" with "farther from 0" (−36 vs 22).

SESSION 2 · TEACHER NOTES

Absolute Value as Total Distance

TALK PROMPTS

- Why does zero sitting between the two readings change how you find the distance?
- Why do you join the two distances instead of taking one away from the other?

PARTNER TASK

Peak and Trench. One partner finds the absolute values of −4.7 and 1.8; the other traces the trip on a number line and finds the total distance (6.5 km). Switch roles with a Drone Climb from 8 m to 23 m (15 m) and compare the two rules.

WATCH FOR

- Subtracting absolute values when 0 is between the readings (−9 to 24 written as 15).
- Adding absolute values when both readings are on the same side of 0 (−18 to −41 written as 59).
- Giving a negative distance.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.NS.C.7c	1	2	2	1	1	2	2	3	2	1	1
S1 · Form B	6.NS.C.7c	1	2	2	1	1	2	2	3	2	1	1
S2 · Form A	6.NS.C.7c	1	2	1	1	2	2	2	3	2	1	1
S2 · Form B	6.NS.C.7c	1	2	1	1	2	2	2	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

absolute value: How many units a number is from **0**.

distance: How many units apart two points are. It is never **negative**.

magnitude: How big a number is, ignoring its **sign**.

- Absolute value is never **negative**, because distance is never negative.
- Closer to 0 means a **smaller** absolute value.

Example: −13 is **13** units left of 0, so |−13| = 13. 6 is 6 units right of 0, so |6| = **6**. Answer: 13 > 6, so **−13** is farther from 0.

YOU TRY

1. a. 21 b. 4.5

2. −2.5, because |−2.5| = 2.5 and 2.5 < 3

EXIT TICKET

1. a. 11 b. 7.25

2. the diver, because |−12| = 12 and 12 > 9

3. **A** (−15)

PRACTICE FORM A

1. a. 16 b. 9 c. 0 d. 2.75

2. a. 6, 2.5, 0.5, 8 b. closest: C (−0.5); farthest: D (8)

3. a. 45 and 12 b. the 12-dollar account

4. 7.5 and 7.5; no, they are on opposite sides of 0

5. 14; the reading is 14 degrees away from 0

6. −36, because |−36| = 36 and 36 > 22

7. No; |−5.2| = 5.2. Absolute value is a distance, and a distance is never negative.

8. a. 85, 40, 120 b. −120 c. 40; greatest means farthest right, not farthest from 0

9. Closeness depends on distance, not on which number is less. |−19| = 19 and |6| = 6. Since 6 < 19, the number 6 is closer to 0.

10. **B** (17)

11. **C** (−3)

Self-check word **MEASURE**

PRACTICE FORM B

1. a. 19 b. 12 c. 0 d. 3.8

2. a. 4, 1.5, 7, 0.5 b. closest: D (0.5); farthest: C (−7)

3. a. 30 and 18 b. the 18-dollar account

4. 6.5 and 6.5; no, they are on opposite sides of 0

5. 9; the reading is 9 degrees away from 0

6. 31, because |31| = 31 and 31 > 27

7. No; |−8.4| = 8.4. Absolute value is a distance, and a distance is never negative.

8. a. 95, 60, 150 b. −150 c. 60; greatest means farthest right, not farthest from 0

9. Tessa confused "less than" with "closer to 0." |−6| = 6 and |4| = 4. Since 4 < 6, the number 4 is closer to 0.

10. **B** (8.5)

11. **C** (−2)

Self-check word **NUMERAL**

SESSION 2 · ANSWERS

Absolute Value as Total Distance

GUIDED NOTES

absolute value: How many units a number is from **0**.

distance: How many units **apart** two points are.

magnitude: The **size** of a number, ignoring its sign.

- If two numbers are on **opposite** sides of 0, add their absolute values.
- If they are on the **same** side of 0, subtract the smaller absolute value from the larger.

Example: 0 is between them, so they are on **opposite** sides. |−8| = 8 and |3| = **3**. Answer: 8 + 3 = **11** blocks

YOU TRY

1. a. 14 b. 8

2. 30 − 12 = 18 m

EXIT TICKET

1. a. 17 b. 11

2. 5 + 13 = 18 degrees

3. **B** (7)

PRACTICE FORM A

1. a. 22 b. 16 c. 15 d. 10

2. a. 12.5 and 7.5 b. 12.5 + 7.5 = 20 ft

3. 4 + 11 = 15 floors

4. 19 − 6 = 13 m

5. 2.4 + 3.6 = 6 km

6. 41 − 18 = 23 m; both readings are on the same side of 0, so subtract

7. 67 + 33 = 100 dollars

8. a. 120 + 340 = 460 m b. 340 − 90 = 250 m c. Trail 1 starts below 0, so it climbs 120 m just to reach 0.

9. 0 is between −9 and 24, so the readings are on opposite sides. Hugo skipped the 9 degrees from −9 up to 0. Add: 9 + 24 = 33 degrees.

10. **B** (15)

11. **B** (add their absolute values)

Self-check word **DISTANCE**

PRACTICE FORM B

1. a. 23 b. 18 c. 14 d. 11

2. a. 10.5 and 7.5 b. 10.5 + 7.5 = 18 degrees

3. 6 + 8 = 14 floors

4. 23 − 8 = 15 m

5. 1.8 + 4.7 = 6.5 km

6. 47 − 12 = 35 m; both readings are on the same side of 0, so subtract

7. 48 + 27 = 75 dollars

8. a. 150 + 280 = 430 m b. 280 − 60 = 220 m c. Trail 1 starts below 0, so it climbs 150 m just to reach 0.

9. 0 is between −7 and 10, so the floors are on opposite sides. Malik should add the distances: 7 + 10 = 17 floors.

10. **B** (15)

11. **C** (10)

Self-check word **ABSOLUTE**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · ABSOLUTE VALUE AS DISTANCE

1. a. 14 b. 8 c. 0 d. 2.5 e. 3/4 f. 1 1/2

2. a. |A| = |−7| = 7 b. |B| = |5| = 5 c. point A

3. a. > (9 > 7) b. = (4 = 4) c. < (−10 < 3) d. = (6.5 = 6.5) e. < (12 < 15) f. < (0 < 1)

4. a. |−35| = 35 and |20| = 20 b. Rosa's (35 > 20)

5. −2, −1, 0, 1, 2 (each is less than 3 units from 0)

6. 4.5 and −4.5. Both are 4.5 units from 0, on different sides.

SHEET 2 · ABSOLUTE VALUE AS TOTAL DISTANCE

1. a. 5 + 6 = 11 b. 1 + 6 = 7 c. 5 − 1 = 4

2. a. 10 b. 8 c. 6.5

3. 8 + 15 = 23 degrees

4. −3 and 9 are on opposite sides of 0: 3 + 9 = 12 floors

5. −9 and 5 are on opposite sides of 0, so he should add: 9 + 5 = 14 units.

6. **B** (19)

7. **B** (38 feet)

October 2026 conceptual additions — teacher key

Session 1 · Absolute Value as Distance

Opposite 9 and absolute value 9. Four is greater than −9, but −9 is farther from zero because its distance is 9 versus 4.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Opposite 9 and absolute value 9. Four is greater than −9, but −9 is farther from zero because its distance is 9 versus 4.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Absolute Value as Total Distance

Across zero: 6+4=10. Same side: 6−2=4. The second segment does not include the trips from both endpoints to zero.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Across zero: 6+4=10. Same side: 6−2=4. The second segment does not include the trips from both endpoints to zero.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.